IS43/IS45(O)

USN	1	M	S							

RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – AUGUST 2024

Program	:	**B.E. – Information Science and Engineering**	**Semester**	:	**IV**
Course Name	:	**Design and Analysis of Algorithms**	**Max. Marks**	:	**100**
Course Code	:	**IS43 / IS45(O)**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Explain with a neat diagram the fundamental steps involved in design and analysis of algorithms.	CO1	(08)
	b)	Write an algorithm to check if all the elements of an array is unique. Analyze its time complexity.	CO1	(06)
	c)	Give a general plan for analyzing time efficiency of recursive algorithms, and analyze the time complexity for tower of Hanoi Puzzle.	CO	(06)
2.	a)	Explain the following notations (formal definition) O, Θ and Ω with an example for each.	CO1	(08)
	b)	Consider the following algorithm: Algorithm Fun(n) //Input: A positive decimal integer n If n==1 return 1 else return Fun(⌊ n/2⌋) +1 i. What does this algorithm compute? ii. What is its basic operation? iii. How many times basic operation is executed? iv. What is the efficiency class of this algorithm?	CO1	(08)
	c)	Compare order of growth algorithm time class 1, n, n^2, n3, 2^n, n!, $\log_2 n$, $n\log_2 n$.	CO1	(04)

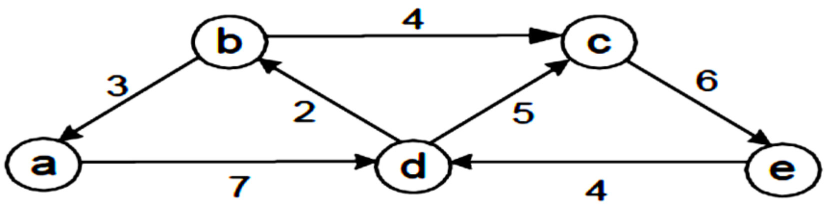
UNIT - II

3.	a)	Illustrate, the tracing of merge sort algorithm for the following set of numbers: 11, 6, 3, 24, 46, 22, 7.	CO2	(05)
	b)	Illustrate, the tracing of bubble sort algorithm for the following set of numbers: 1,16, 30, 4, 6, 2, 27.	CO2	(05)
	c)	Write a quick sort algorithm and analyze its time complexity for the worst and best case efficiency.	CO2	(10)
4.	a)	Apply Horspool algorithm to determine the number of character comparisons made for searching the pattern "ROB" in the text: "LIFE IS FULL OF PROBLEM". Write the horspool algorithm.	CO2	(08)
	b)	Write an algorithm to sort the elements using quick sort method. Trace quick sort algorithm for the given array of numbers, also show the tree calls and compute average time complexity. 2, 7, 8, 3, 1, 9, 5, 6, 3	CO2	(08)
	c)	Define brute force approach. Indicate why it is an important algorithm design strategy.	CO2	(04)

UNIT - III

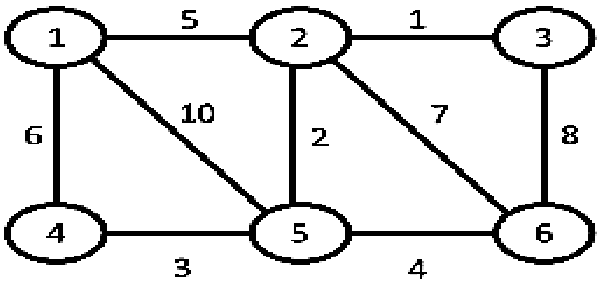
5. a) Solve the following instance of Knapsack problem using the bottom up dynamic programming technique to obtain: Optimal subset of items with the given data N=5, [w1,w2,w3,w4,w5]=[10,20,30,40,50] and [p1,p2,p3,p4,p5]=[20,30,66,40,60] and M=100. Give the time complexity of Knapsack problem. — CO3 (07)

b) State and explain Dijikstra's algorithm to find single source shortest path for a given graph. Apply the same for the following graph. Assume "a" as the source vertex. — CO3 (08)



c) Write Kruskal's algorithm to construct minimum cost spanning tree. — CO3 (05)

6. a) Explain how dynamic programming is used to compute all pair's shortest paths for a weighted digraph, and apply the same to compute all pair shortest paths for the following graph. — CO3 (08)



b) i) Construct a Huffman code for the following data: — CO3 (08)

character	A	B	C	D	E
probability	0.4	0.1	0.2	0.14	0.16

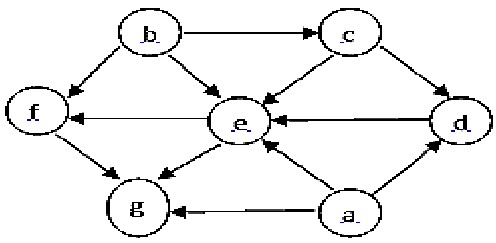
ii) Encode the text ABACABAD using the code.
iii) Decode the text whose encoding is 100010111001010 using Huffman code.

c) Compare greedy and dynamic programming techniques. — CO3 (04)

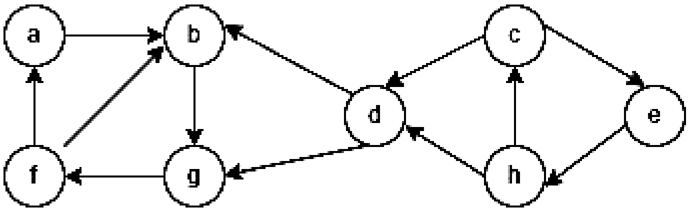
UNIT- IV

7. a) Apply insertion sort for the following instance of an array [50,30,10,40,20,80,90,60]. Compute its worst case time complexity. — CO4 (08)

b) Design an algorithm for Breadth-First-search traversal. Determine the BFS for the following and write topological sorting for the following using source removal method. — CO4 (12)



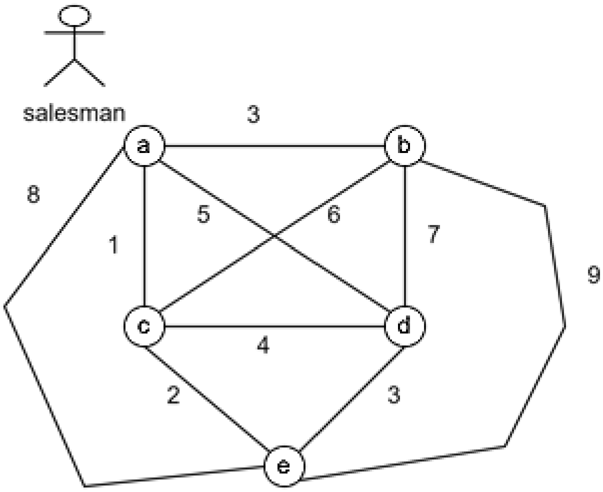
8.	a)	Differentiate between DFS and BFS.	CO4	(05)
	b)	What is topological sorting? Arrange the following graph nodes in topological sorting using DFS.	CO4	(05)



	c)	What are AVL trees? Discuss four different rotations with respect to AVL tree. Construct AVL tree for the following list of elements: 3, 5, 11, 8, 4, 1, 12, 7, 2, 6, 10.	CO4	(10)

UNIT- V

9.	a)	What is a heap? Construct a heap for the list 1, 8, 6, 5, 3, 7, 4 by bottom –up algorithm. And sort the given list by using by heapsort.	CO5	(10)
	b)	What is backtracking? Explain how backtracking can be used to solve n-Queen's problem and obtain one solution to 4-Queen's problem showing the state Space tree.	CO5	(10)
10.	a)	Apply branch and bound algorithm to solve the travelling salesman problem for the given graph.	CO5	(10)



	b)	Explain the following with an example: (i) P-Problem (ii) NP-Problem (iii) NP-Complete Problem (iv) NP-Hard.	CO5	(10)

****